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Patent Search

Invention Title	A MULTIMODAL HEALTH MONITORING SYSTEM VIA SMART MIRROR AND WEARABLE SYNCHRONIZATION
Publication Number	41/2025
Publication Date	10/10/2025
Publication Type	INA
Application Number	202511089243
Application Filing Date	18/09/2025
Priority Number	
Priority Country	
Priority Date	
Field Of Invention	BIO-MEDICAL ENGINEERING
Classification (IPC)	A61B0005000000, A61B0005110000, A61B0005020500, G06F0003160000, G06F0003010000

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Abstract:

The present invention relates to a multimodal health monitoring system(100) via smart mirror and wearable synchronization. The present invention includes a smart unit(102), a smartwatch(126) and a power adapter(128). The smart mirror unit(102) includes a two-way mirror glass(104), a display screen(106), a camera(108), a motion sensor(110), a temperature sensor(112), a microphone(114), a speaker(116), a wireless module(118) and a computing unit(120). The computing unit(120) is connected way mirror glass(104), the display screen(106), the camera(108), the motion sensor(110), the temperature sensor(112), the microphone(114), the speaker(116) and the module(118) and includes a memory unit(122) and a processing unit(124). The smartwatch(126) is connected wirelessly to the smart mirror unit(102) through the wireless module(118) for monitoring health statistics and physical activity of user and send monitored data to the smart mirror unit(102). The power adapter(128) is connected to the smart mirror unit(102) and provides constant power supply.

Complete Specification

health indicators such as heart rate, respiratory rate, skin condition, body posture, and emotional state, while maintaining an intuitive, non-disruptive user experience.

OBJECTIVE OF THE INVENTION

The main objective of the present invention is to develop an artificial intelligence based multimodal health monitoring system.

Yet another objective of the present invention is to synchronize a wearable smartwatch and a smart mirror into a single interactive platform for health monitoring.

Yet another objective of the present invention is to develop facial based wellness detection using image processing.

Yet another objective of the present invention is to monitor health statistics data in real time, offering personalized views and suggestions.

Yet another objective of the present invention is to implement hands free operation with visual feedback by integrating non-touch, camera and sensor-based interaction allowing user to receive health updates.

SUMMARY OF THE INVENTION

The present invention relates to a multimodal health monitoring system via smart mirror and wearable synchronization. The present invention includes a smart mirror unit, a smartwatch and a power adapter. The smart mirror unit includes a two-way mirror glass, a display screen, a camera, a motion sensor, a temperature sensor, a microphone, a speaker, a wireless module and a computing unit. The two-way mirror glass enables screen to be visible behind reflective surface, functioning both as a mirror and display. The display screen is positioned behind the two-way mirror glass and configured to display time, notifications, health statistics data, daily health reminders, weather updates, calendar and yoga instructions. The display screen is selected from LCD or LED. The camera is used for facial recognition and monitoring yoga postures. The camera utilizes OpenCV or Media Pine for facial recognition and monitoring yoga postures. The motion sensor enables touchless control via hand gestures for interaction with the system.

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Page last updated on: 26/06/2019